The publication activity and migration trends of Ukrainian scientists in the social sciences and humanities during the first two years of the Russo-Ukrainian war

Serhii Nazarovets[1]

[1] Borys Grinchenko Kyiv Metropolitan University, 18/2 Bulvarno-Kudriavska Str., 04053 Kyiv, Ukraine; serhii.nazarovets@gmail.com

Abstract

This study analyses the publication activity and migration patterns of Ukrainian scholars in the social sciences and humanities (SSH) during the initial two years of the Russo-Ukrainian war. Focusing on scholars who published at least three papers, the study underscores the resilience of these scholars, who continued their academic endeavours within their homeland despite the conflict. The research utilizes data from the Social Sciences Citation Index (SSCI) and the Arts & Humanities Citation Index (AHCI) to illustrate their continued scientific contributions under adverse conditions. It also highlights the crucial role of international collaboration in supporting Ukrainian SSH research, emphasizing that such collaborations primarily manifest through joint research projects rather than relocation of scholars to foreign institutions.

Keywords: publication trends; migration trends; Russo-Ukrainian war; bibliometrics

1. Introduction

Conducting scientific research in Ukraine has always presented formidable challenges, characterized by slow and inconsistent reforms, underdeveloped international cooperation, inadequate funding, outdated equipment, and a notable gap between the higher education sector and industry (Gaind & Liverpool, 2023; Kozak et al., 2015; OECD, 2022; Schiermeier, 2019; Yegorov, 2018). The recent annexation of Crimea and Russia's hybrid warfare against Ukraine, which escalated into a full-scale Russian invasion in February 2022, has further compounded the difficulties faced by Ukrainian scientists. This war has inflicted significant damage on the Ukrainian and global scientific communities, manifesting in the physical destruction and occupation of scientific institutions, the depletion of research infrastructure and funding, and the loss of scientific personnel due to casualties, injuries, and forced migration (da Costa et al., 2022; Irwin, 2023; Oleksiyenko et al., 2021; Pereira et al., 2022; Stone, 2015).

The academic community has promptly responded to this significant military conflict, the largest in Europe since 1945, by universally condemning Russia's aggression against Ukraine. Numerous academic institutions worldwide have extended their support to Ukrainian scientists by offering them temporary work invitations (Gaind et al., 2022; Maryl et al., 2022; Rose et al., 2023). Additionally, scientific journals have actively engaged in publishing articles that scrutinize the adverse impacts of the Russian invasion on Ukrainian science. These articles explore overarching issues related to the organization of scientific endeavours during wartime, addressing both the institutional and individual challenges faced by researchers.

Following February 2022, the productivity of Ukrainian scientists may have been adversely affected by the ongoing war. These scientists could have tragically lost their lives, been captured, conscripted into the military, transitioned to volunteer work, encountered restricted access to equipment and materials, or simply did not maintain scientific pursuits at their accustomed level due to heightened stress and anxiety (Karatzias et al., 2023; Tsybuliak et al., 2023; Xu et al., 2023). A study by Lushchak et al. (2024) revealed that forced displacement from one's prior place of residence can significantly exacerbate mental health challenges stemming from the impacts of war. Consequently, even Ukrainian research migrants who have secured access to their partners' research infrastructure may confront substantial obstacles in integrating into the new academic milieu.

A significant portion of Ukrainian scientists have been compelled to emigrate and pursue employment opportunities both domestically (Suchikova et al., 2023) and abroad (Karmadonova, 2023; Kaczmarczyk, 2022; Lipták & Kincses, 2023). According to a survey conducted by de Rassenfosse et al. (2023), approximately 18.5% of Ukrainian academics have left the country. A comprehensive survey conducted by the UAScience.Reload initiative in April 2022, involving 2,173 Ukrainian scientists, revealed that the ongoing conflict has drastically affected their professional engagement and financial stability, with over 72% unable to continue their research activities as before, and a significant majority experiencing worsened economic conditions (Lutsenko et al., 2023). However, prior studies have predominantly focused on employees of research institutes and universities in general, with no efforts made to assess the potential threat posed by the departure of productive researchers in specific research fields.

In their analysis of the consequences of the Russian invasion on Ukrainian science, Ganguli & Waldinger (2023) demonstrate a significant disruption in scientific activities, including a 10% drop in research outputs and substantial damage to physical infrastructure at leading universities. The study further highlights the displacement of prominent scientists and a drastic reduction in international collaborations, particularly with Russian institutions, posing long-term challenges to the Ukrainian scientific community. However, it is important to note that the research method employed by Ganguli & Waldinger, which primarily focused on the number of publications, likely resulted in a sample that under-represents Ukrainian scholars in the social sciences and humanities.

This study examines the changes in the publication behaviour of the productive Ukrainian scientists working in SSH during the initial two years of the Russo-Ukrainian war, with a focus driven by several significant considerations. It is widely acknowledged that history and culture can profoundly influence the trajectory of SSH development (Kulczycki et al., 2018). Following Ukraine's independence in 1991, Ukrainian SSH fields were liberated from the constraints of Soviet ideology and censorship. However, the ongoing war poses a threat to the progress made in these scientific domains within Ukraine. Moreover, researchers in SSH are often more focused on local issues than in the natural sciences (Nederhof, 2006). Consequently, their research may be less dependent on direct physical participation in international scientific events. It is conceivable that Ukrainian SSH scientists have managed to sustain productive work amid the challenges of war, provided with basic physical security and minimal funding.

This study examines the subset of Ukrainian scholars in SSH who are distinguished by their prolific publication records relative to their peers. Highly productive scientists represent a small fraction of the total number of scientists in a country. Nonetheless, they play a pivotal role in shaping the research landscape and influencing the impact of research conducted within the country, both presently and in the future. This assertion is supported in studies conducted by Abramo et al. (2017), Fox & Nikivincze (2021), and Kwiek & Roszka (2023). According to Kwiek (2016), the most productive 10% of scientists contribute to 50% of scientific papers. Moreover, productive scientists not only publish their own work but also serve as mentors, sharing their knowledge and experience to help elevate other researchers within the country to international acclaim (Agrawal et al., 2017; Yin & Zhi, 2017). Therefore, the potential loss of Ukraine's productive scientists could have catastrophic consequences in the wake of the Russo-Ukrainian war. Nevertheless, the urgency of addressing the impact of the war on productive Ukrainian scientists remains unacknowledged. However, it is important to contextualize these insights within the Ukrainian scenario: the publication activity of Ukrainian scholars in leading international journals is relatively limited, and this study focuses on those who are productive within the specific confines of Ukraine's research environment. The criteria and rationale for selecting productive Ukrainian scientists in SSH are described in the Data and methods section.

The inquiry into whether there have been notable shifts in the percentage of internationally published works by the most productive Ukrainian scientists in SSH is of significant interest. Since the onset of the war, international scientists have displayed a heightened willingness to engage Ukrainian partners in academic collaboration (Oleksiyenko et al., 2023). This inclination should theoretically lay the groundwork for an uptick in the international activity of Ukrainian scientists. However, Ukrainian scientists may also encounter circumstances beyond their control that impede their ability to respond to proposals from foreign scientific partners. Furthermore, they may face resource constraints hindering the continuation of ongoing international research or find themselves compelled to redirect the focus of their research (Fiialka, 2022). Damaševičius and Zailskaitė-Jakštė (2023) observed a trend suggesting a reduction in the diversity of international collaborations among Ukrainian researchers in 2022. Nonetheless, it is crucial to note that these findings were derived from data sourced from a single academic publisher.

Another crucial aspect is to evaluate the significance of international scientific collaborations in the work and migration dynamics of Ukrainian scientists (Korennoy & Khorevin, 1995). It can be inferred that if a Ukrainian scientist had been actively engaged in collaborations with foreign counterparts prior to the onset of the Russo-Ukrainian war, then in such instances, foreign colleagues might be more inclined to extend offers for Ukrainian scientists to work in their respective countries. Concurrently, it may have been relatively easier for Ukrainian scientists to contemplate relocating to other countries under such circumstances.

It is significant to investigate the migration patterns of productive Ukrainian scientists during the war, with particular consideration of gender and age dynamics. Notably, women in Ukrainian science demonstrate higher publication productivity compared to men, a trend often observed in developing countries with limited scientific output (Larivière et al., 2013). Moreover, Ukrainian legislation restricts men aged between 18 and 60 who are mandated to serve in the military are restricted from travelling abroad during the enforcement of martial

law in Ukraine[1]. Consequently, it is anticipated that female scientists will exhibit a higher migration rate than their male counterparts. However, the impact of this phenomenon on the productivity of Ukrainian scientists remains unclear.

Under martial law, Ukrainian men require a special permit even for short-term visits abroad, which makes it difficult for them to participate in international conferences and other scientific events, with potentially negative consequences for scientific productivity. However, Ukrainian scientists have been granted a deferral from mobilization, affording them the choice between military service and the continuation of their research. Consequently, despite encountering several adverse factors, the publishing activity of male scientists may not have been significantly impacted at a national level.

This study endeavours to answer two pivotal questions about the impact of the Russo-Ukrainian war on the academic sector within the social sciences and humanities:

Q1: Has the Russo-Ukrainian war affected the total number of the most productive Ukrainian scientists in SSH, who continue to operate within the country? This question also explores the age and gender characteristics of these scientists and examines whether there have been changes in the structure of their collaborations with foreign co-authors since the war began.

Q2: What is the extent of emigration among these most productive Ukrainian scientists since the onset of the war? Moreover, how many of them are currently engaged in research activities at foreign institutions?

This paper is structured as follows: The "Data and Methods" section provides detailed information on the sources and characteristics of the dataset used in this study, along with an explanation of the rationale for selecting specific sampling criteria. The "Results" section concisely presents the key findings. Finally, the "Discussion and Conclusions" section summarizes the study's insights and discusses its limitations.

2. Data and methods

Ukraine has long grappled with an imperfect system for evaluating academic productivity, which predominantly relies on quantitative metrics of research activity (Abramo et al., 2023). Consequently, there is a propensity among Ukrainian authors to prioritize quantity over quality, with the aim of publishing as many papers as possible in journals indexed in relevant citation databases (Hladchenko, 2022; Hladchenko, 2023). This flawed national assessment framework contributes to a surge in publications by Ukrainian authors, particularly in local journals (Hladchenko & Moed, 2021; Nazarovets, 2020; Nazarovets & Mryglod, 2023), as well as in journals that have been delisted from citation databases, often due to breaches of research ethics (Nazarovets, 2022). Additionally, there is a notable rise in excessive self-citation within these publications (Ioannidis et al., 2019; Baccini & Petrovich, 2023).

In order to minimise the inclusion of authors of non-peer-reviewed papers in the sample for this study, highly selective lists of journals, namely two WoS CC indexes – the Social

[1] https://zakon.rada.gov.ua/laws/show/57-95-%D0%BF#Text

Sciences Citation Index (SSCI) and the Arts & Humanities Citation Index (AHCI) – were utilized in the initial stage of papers and authors selection. The editors of the SSCI and AHCI adhere to stringent content selection criteria in order to ensure that these indexes encompass the most authoritative journals in the SSH[2]. While this approach, of relying on selective journal lists, has its limitations concerning the comprehensiveness of research field representation (Pölönen et al., 2021), it ensures the inclusion of journal papers that have undergone proper peer review for this analysis. Therefore, to provide a broad quantitative overview of the publication activity of Ukrainian scientists in the SSH, the first phase of the study involved searching the SSCI and AHCI for publications within the last 10 years in which at least one author indicated their affiliation with a Ukrainian institution (CU=Ukraine). This search was conducted on 26 April 2024, with only articles and reviews being considered in the search results.

For the subsequent stage of this study, the search was limited to the years 2021 (the last year preceding the war) and 2023 (the second year of the full-scale Russo-Ukrainian war). The general quantitative attributes of Ukrainian scientists papers in SSH in these two distinct years were delineated, and a comparative analysis was conducted to discern changes in the publication activity of Ukrainian authors during this period. The examination of the author composition of these papers facilitated the identification of authors with the highest publication count in a given year. In order to identify the most prolific authors in the SSH fields, this study focused on Ukrainian scholars who had published at least three articles in a single year. This threshold was established based on the observation that Ukrainian scientists typically contribute a limited number of articles to influential journals, resulting in a relatively small cohort of scholars with publication rates exceeding three papers per year. For instance, data from the SSCI and the AHCI reveal that in 2021, Ukrainian scientists produced fewer than 400 articles and reviews. In contrast, scientists from Poland, which shares a border with Ukraine and has a comparable population size, published nearly 7,000 such works in the same year. The decision to exclude Ukrainian scholars who published only one or two papers per year was informed by the objective to analyse the productivity of the most active contributors within SSH fields, because precisely scholars with higher publication rates are often having greater influence on the development of their disciplines.

Conducting quantitative studies on the international migration of researchers poses challenges due to the dearth of reliable and up-to-date data. Consequently, such studies often rely on tracking researchers' movements by monitoring changes in the affiliations indicated in their papers (Moed & Halevi, 2014; Poitras & Larivière, 2023; Robinson-Garcia et al., 2019). This method was employed in the current study as well – initially, the affiliations of the most productive Ukrainian scientists in 2021 were established, followed by an analysis to ascertain if there were any changes in their institutional affiliations in 2023. Given that many scientists may have submitted their manuscripts in 2021, their papers published in 2022 may not yet reflect updated affiliations. Therefore, a comparison of the publications of these scientists in 2021 with their publications in 2023 was deemed appropriate. By 2023, authors would have had sufficient time to indicate their current affiliations, ensuring a more accurate assessment of migration patterns and changes in institutional affiliations.

[2] https://clarivate.com/products/scientific-and-academic-research/research-discovery-and-workflow-solutions/webofscience-platform/web-of-science-core-collection/editorial-selection-process/editorial-selection-process/

The SSCI and the AHCI do not encompass the entirety of scientists' publication activity. In order to gain a more comprehensive understanding of the publication behaviour of the analysed authors in the relevant years, I also determined the total number of articles and reviews authored by them in all journals indexed in the Web of Science Core Collection (WoS CC). This was achieved by using the Web of Science ResearcherID of these scientists. The limited number of publications by Ukrainian scholars in these fields over the years allowed for a manual analysis of bibliographic data to ascertain gender, age, and verify information regarding the current place of work of the researcher. To accomplish this, I manually supplemented the WoS CC data with information sourced from the official websites of the institutions and the personal pages of the researchers on specialized academic platforms such as ORCID and Google Scholar Citations.

In order to verify the accuracy of the listed affiliations of the selected Ukrainian scientists – considering the possibility of outdated information or instances where scientists might have maintained a Ukrainian institutional affiliation despite residing abroad – I contacted each scientist directly. Emails were sent to confirm their presence in Ukraine in 2023 and to ensure that their research papers were genuinely authored within the country. Additionally, I invited respondents to provide qualitative insights into their experiences of conducting scientific work during the Russo-Ukrainian war. I encouraged them to share the factors that enabled and motivated them to continue their research activities under challenging circumstances. This approach was designed to gain a deeper understanding of both the personal and professional impacts of the war on their scientific work.

3. Results

3.1 Bibliometric analysis

Between 2014 and 2023, Ukrainian scientists contributed a total of 2,504 articles and reviews to journals indexed in the SSCI and AHCI. Notably, the number of publications exhibited a consistent increase each year, despite the backdrop of the Russo-Ukrainian war (Fig. 1). Table 1[3] presents the Ukrainian scientists affiliated with Ukrainian institutions who had at least three publications in SSCI and AHCI in 2021. A total of 22 productive researchers (11 women and 11 men) were identified in 2021. It should be noted that these authors are productive scientists within the Ukrainian SSH, but not necessarily within the global SSH. It is crucial to clarify that the method outlined in this paper for selecting scientists via SSCI and AHCI was solely used to identify Ukrainian scientists who disseminate their research findings in influential scientific journals. The lists of scientists provided should not be construed as a ranking of scientists. To prevent any potential misinterpretation, the names of the scientists in Tables 1 and 2 are presented in alphabetical order.

In 2023, the publication activity of 16 of these researchers experienced a decrease, while 2 remained unchanged, and 4 researchers showed an increase compared to their output in 2021. Information regarding the research activity and current affiliation of one individual could not be ascertained. Notably, none of the identified researchers (21) ceased indicating

[3] All data presented in the tables 1 and 2 are sourced from Web of Science databases. The information is non-confidential and available for public review.

a Ukrainian institution in their publications. Furthermore, in 2023, four researchers (3 men and 1 woman) disclosed affiliations with foreign institutions in addition to their Ukrainian affiliations. According to the Web of Science classification, Ukrainian authors demonstrated the highest productivity in the following research areas in 2021: environmental sciences and ecology (81 publications), public environmental occupational health (46), science technology (other topics) (46), and business economics (30). Collaboration with foreign partners played a pivotal role for the majority of productive Ukrainian researchers in the field of SSH. Indeed, 72% of their publications were the result of international collaboration, while national collaborations and individual works each accounted for 14%. Noteworthy international collaborations of Ukrainian scientists in 2021 were observed with Poland (84), the USA (72), England (59), Germany (53), and Italy (44).

In 2023, a total of 32 productive Ukrainian scientists were identified, comprising 13 women and 19 men. Table 2 presents the Ukrainian scientists affiliated with Ukrainian institutions who had at least three publications in SSCI and AHCI in 2023. Remarkably, despite the ongoing war, 25 of these scientists managed to increase their annual publication activity compared to 2021. Additionally, besides their affiliation with Ukrainian institutions, 8 authors (4 men and 4 women) indicated affiliations with foreign institutions in 2023. Concurrently, one author had previously indicated affiliations with two institutions prior to the outbreak of the war, while another author commenced indicating a Ukrainian institution during the war. The remaining five authors solely indicated Ukrainian affiliation in their articles before the war.

In 2023, the most productive Ukrainian authors were in the following research areas according to Web of Science classification: environmental sciences and ecology – 99, science technology (other topics) – 70, psychology – 51, business economics – 49, public environmental occupational health – 46. Notably, Ukrainian scientists continued to produce a significant percentage of papers through international collaborations, accounting for 71%. Additionally, there was a modest increase in national collaborations, comprising 18%, while the proportion of single-authored publications decreased to 11%. The most productive international collaborations for Ukrainian scientists in 2023 were with Poland (109), the USA (82), China (52), Germany (48), England (47), and Italy (35). Hence, throughout the initial two years of the full-scale war, the structure of Ukraine's most productive scientific collaborations remained largely unchanged. The only noteworthy alteration was the inclusion of China in these top collaborations in 2023.

In summary, this study identified a total of 47 productive Ukrainian authors in the field of SSH for the years 2021 and 2023, encompassing 20 women and 27 men. Notably, 64% of these researchers were able to maintain or enhance their annual publication activity in journals indexed in WoS CC. Furthermore, only 11 (23%) of the researchers indicated additional foreign affiliations in their papers, alongside their Ukrainian institution. Specifically, 6 men and 5 women included such foreign affiliations in their papers. Importantly, none of the analysed authors relinquished their Ukrainian affiliation. One author altered their Ukrainian affiliation, while information regarding the subsequent research career of another productive author from 2021 could not be obtained. Additionally, the majority of researchers (77%) were exclusively engaged in a single Ukrainian institution during the period from 2021 to 2023. Notably, all male scientists included in this study were under the age of 60.

3.2 Analysis of online survey

A total of 46 emails were dispatched to a selected group of Ukrainian scientists, requesting information about their country of residence during their research activities in 2023. A response was received from 19 scientists, representing a response rate of 41%. All but one of the responses validated the location information provided in their respective publications. Specifically, 15 respondents confirmed that they had conducted their research within Ukraine, while four respondents, though working abroad, indicated that they had maintained active scientific collaborations with an institution in Ukraine.

In his answers to questions about the peculiarities of conducting scientific research in Ukraine during the war, several respondents highlighted an increase in international collaboration. Foreign scientists and institutions have actively sought partnerships with Ukrainian researchers, often providing essential resources and invitations to participate in international projects. This surge in support not only aids in maintaining ongoing research but also enhances the global visibility of Ukrainian scientific contributions. There is a notable increase in feedback and recognition from international peers, which has boosted morale and spurred further engagement in research activities. For many, this external validation has been crucial during times of uncertainty and stress. Financial support from international bodies and the waiving of article processing charges (APCs) by journals also have alleviated some of the financial burdens on Ukrainian scientists, allowing them to focus on their research without the additional stress of funding.

Emotional resilience plays a significant role in the continuation of scientific work. Several scientists described their research as a form of sublimation – a way to distract from the harsh realities of war and to maintain mental health. The sense of duty towards students and the scientific community also emerged as a strong motivating factor, driving researchers to work with heightened determination under difficult circumstances.

The personal accounts from the scientists provide a profound insight into their resilience. One researcher stated, "My team is here, and only by my example can I demonstrate the strength of the spirit, the desire to teach students and the opportunities to develop in Ukraine." Another highlighted the extraordinary circumstances, noting, "Among the illustrative examples, I can mention the work of colleagues who have been fighting on the front lines since the first days of the war, but in some incredible way find the opportunity to work on rotation." This commitment extends to the broader mission of science, as another scientist passionately declared, "Our mission and our passion is to do science. And whether they bomb us or want to destroy us, we will defend our culture and the right to intellectual sovereignty."

Discussion and conclusions

In this paper, through a quantitative analysis of publication activity, the aim was to ascertain whether there were any changes in the number and publication behaviour of the most productive Ukrainian scientists in SSH during the initial two years of the Russo-Ukrainian war. The findings of the study suggest that despite the enduring conflict, Ukrainian scientists, alongside officials and other scientific stakeholders, managed to sustain scientific endeavours within the country. This resilience was facilitated, at least in part, by support

from foreign partners. Contrary to pessimistic projections, there was no notable decline observed in the quantity of Ukrainian publications in the field of SSH. The question remains that, if not for the war, the growth of Ukrainian research papers could have been much greater, and it is also likely that certain negative effects will become noticeable only in a few years.

Although not all productive authors identified for 2021 were able to maintain their high publication activity in subsequent years, in only one case out of 47 analysed was there no information on the author's continuing research activity. Thereby, it is evident that Ukrainian scientists in SSH persist in their research pursuits. While the numerical significance of these scholars and their scientific outputs may appear modest in the global scale, it underscores their critical importance for the future research trajectory of Ukraine. Since achieving independence, Ukraine has steadily enhanced its contribution to the global treasury of knowledge in SSH. These scientists, along with their research work within Ukrainian institutions, play a pivotal role in disseminating and imparting knowledge to forthcoming generations of Ukrainian scientists in these fields (Fisher, 2023).

Ukrainian productive scientists in SSH persist in their work within Ukrainian institutions, with nearly a quarter of them indicating additional foreign affiliations during the course of the war. Interestingly, my analysis did not uncover any significant gender disparities in this regard. Notably, all male scientists included in this study were under the age of 60, rendering them theoretically subject to border crossing restrictions, which could potentially impede the establishment of international scientific partnerships. However, Ukrainian legislation provides male scientists with a deferral from mobilization, enabling them to sustain their research activities effectively, as evidenced by the findings of this analysis. Furthermore, the example of Ukrainian female scientists, who are exempt from military service, indicates that the opportunity for border crossing has not precipitated widespread emigration abroad among this demographic.

Presently, Ukrainian government officials frequently express their aspiration to repatriate Ukrainian academics who were compelled to depart the country following February 24, 2022[4]. However, the findings of this study underscore that the most productive Ukrainian scientists in SSH have opted to remain at the institutions where they were employed prior to the Russian invasion.

It is noteworthy that approximately three-quarters of the papers authored by productive Ukrainian scientists involve collaboration with foreign researchers, underscoring the paramount importance of international collaboration and maintaining ties with foreign partners. Amid the quarantine restrictions imposed due to the COVID-19 pandemic, many research teams gained valuable experience in fostering international scientific cooperation online (Chacón-Labella et al., 2021; Rudzin et al., 2022). It is conceivable that Ukrainian scientists were able to leverage this experience effectively during the war, although empirical evidence is required to substantiate this hypothesis.

During the initial two years of the war, the structure of international research collaborations among Ukrainian scientists remained largely unchanged, suggesting that the Russo-

[4] For instance https://www.ukrinform.ua/rubric-technology/3784518-ukraina-znacno-zbilsue-vitrati-na-naukovi-rozrobki-lisovij.html

Ukrainian war did not exert a significant impact on established models and trends of international scientific cooperation in SSH. However, it is evident that Ukraine's international scientific partnerships should be expanded. Nevertheless, amidst the Russian military invasion, Ukraine faces challenges in creating the requisite conditions and allocating resources for such expansion.

The results derived from this quantitative analysis did not provide insights into which Ukrainian scientists have availed themselves of offers from foreign partners to work and train at their institutions (Rassenfosse et al., 2023; Rose et al., 2023). Given the focus of this study on productive scientists, who typically possess extensive experience within a particular research domain, it is plausible that such individuals may encounter challenges in relocating their work to another country. Factors such as potential demotion, alterations in living conditions, among others, may not align with their established career trajectories and personal or family circumstances. Consequently, offers from foreign partners may be more appealing to early-career researchers, who are typically more inclined to embrace the associated risks of relocation and are yet to establish high positions and salaries (Khan, 2021). However, it's important to note that early-career researchers, owing to their relatively fewer publications, were not included in this analysis.

Moreover, it is essential to acknowledge that this study is confined to the analysis of SSH, and it's plausible that productive Ukrainian scientists in the natural sciences may exhibit different patterns. Given that quantitative analysis alone may not fully elucidate the underlying reasons for this behaviour, conducting additional interviews with Ukrainian scientists in the future could serve as a valuable avenue to delve deeper into this issue.

This paper focuses solely on the quantitative aspects of publication activity of Ukrainian scholars in SSH. However, delving into the citation patterns of their publications could provide valuable insights into the impact of their work. Nevertheless, given the limited time elapsed for the accumulation of citations, studying the impact of works produced during the Russo-Ukrainian war remains a prospect for future research.

Furthermore, while there may be a greater number of works by Ukrainian scholars available in Web of Science, it is important to acknowledge the known limitations in the completeness of metadata. As highlighted by Mryglod and Nazarovets (2023), not all records may be accurately identified as papers authored by Ukrainian authors due to these limitations. Additionally, the presence of multiple author profiles for scientists in the Web of Science could further contribute to the dispersion and incomplete coverage of their publication activity within this study. Also, it is important to consider the indexing delay inherent in SSCI and AHCI – publications may not appear in the databases immediately after publication, which could affect the accuracy of data analysis. These factors underscore the need for cautious interpretation of the quantitative data presented in this paper.

Ukrainian writer Larysa Kosach metaphorical pondered in her poem, "Mother, does every bird fly away for the winter?". This study's findings resonate with this sentiment, revealing that the most productive Ukrainian scientists in SSH have decided to bravely face and survive the "winter" in their homeland. Despite the slow growth in the publication activity of the Ukrainian SSH community, what remains paramount is the unyielding continuity of the scientific process amidst the backdrop of the Russo-Ukrainian war. This perseverance owes

much to international cooperation and the active engagement of foreign research partners. For productive Ukrainian authors, this support manifests not merely in the form of open positions in foreign institutions, but rather through participation in collaborative and influential research projects. As the war persists, it becomes increasingly evident that Ukrainian scientists will continue to rely on such support from the global academic community to sustain their scholarly pursuits.

Acknowledgements
The author would like to thank all Ukrainian defenders for the possibility to finalize and publish this work.

Competing interests
The author do not have any competing interests.

Funding statement
This study was not supported by any sponsor or funder.

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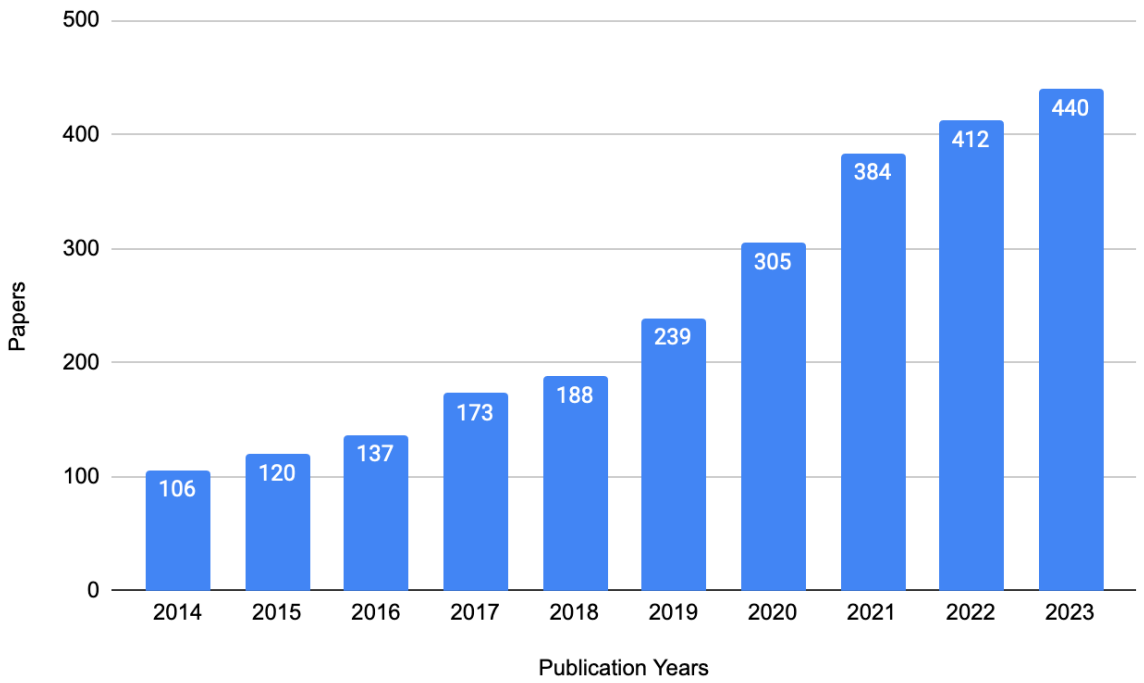

Figure 1. Articles and reviews of Ukrainian scientists published in 2014-2023 in the journals covered by SSCI and AHCI

Table 1. Authors affiliated with Ukrainian institutions with at least three publications in SSCI and AHCI in 2021

Authors	Papers in 2021		Affiliation in 2021	Gender	Papers in 2023		Affiliation in 2023
	SSCI & AHCI	WoS CC			SSCI & AHCI	WoS CC	
Bogdanov, Sergiy	3	6	National University of Kyiv Mohyla Academy, Kyiv, Ukraine	Male	0	2	National University of Kyiv Mohyla Academy, Kyiv, Ukraine
Danyliuk, Ivan	3	9	Taras Shevchenko National University of Kyiv, Kyiv, Ukraine	Male	5	8	Taras Shevchenko National University of Kyiv, Kyiv, Ukraine
Dumchev, Kostyantyn	6	11	Ukrainian Institute on Public Health Policy, Kyiv, Ukraine	Male	0	3	Ukrainian Institute on Public Health Policy, Kyiv, Ukraine
Fedoriak, Mariia	3	5	Yuriy Fedkovych Chernivtsi National University, Chernivtsi, Ukraine	Female	2	2	Yuriy Fedkovych Chernivtsi National University, Chernivtsi, Ukraine
Galkin, Andrii	3	6	O. M. Beketov National University of Urban Economy in Kharkiv, Kharkiv, Ukraine	Male	0	1	O. M. Beketov National University of Urban Economy in Kharkiv, Kharkiv, Ukraine
Kiriazova, Tetiana	6	9	Ukrainian Institute on Public Health Policy, Kyiv, Ukraine	Female	4	4	Ukrainian Institute on Public Health Policy, Kyiv, Ukraine
Korolyov, Igor	4	6	Taras Shevchenko National University of Kyiv, Kyiv, Ukraine	Male	0	0	Taras Shevchenko National University of Kyiv, Kyiv, Ukraine
Kozytska, Inna	4	4	Taras Shevchenko National University of Kyiv, Kyiv, Ukraine	Female	4	4	Taras Shevchenko National University of Kyiv, Ukraine
Kurapov, Anton	5	11	Taras Shevchenko National University of Kyiv, Kyiv, Ukraine	Male	8	8	Taras Shevchenko National University of Kyiv, Kyiv, Ukraine; **University of Salzburg, Austria***
Lozynskyi, Vasyl	3	10	Dnipro University of Technology, Dnipro, Ukraine	Male	0	4	Dnipro University of Technology, Dnipro, Ukraine; **Guangdong University of Petrochemical Technology, Maoming, China**

Lyulyov, Oleksii	3	10	Sumy State University, Sumy, Ukraine	Male	11	22	Sumy State University, Sumy, Ukraine
Mryglod, Olesya	3	3	Institute for Condensed Matter Physics, National Academy of Sciences of Ukraine	Female	0	0	Institute for Condensed Matter Physics, National Academy of Sciences of Ukraine
Nehuliaieva, Liliia	3	3	Andrei Krupynskyi Lviv Medical Academy, Lviv, Ukraine	Female	0	0	*No information*
Niemets, Liudmyla	3	6	V.N. Karazin Kharkiv National University, Kharkiv, Ukraine	Female	0	0	V.N. Karazin Kharkiv National University, Kharkiv, Ukraine
Pavlova, Daria	3	3	Ukrainian Institute for Social Research after Olexander Yaremenko, Kyiv, Ukraine	Female	0	1	Ukrainian Institute for Social Research after Olexander Yaremenko, Kyiv, Ukraine
Pavlova, Iuliia	6	9	Lviv State University of Physical Culture, Lviv, Ukraine	Female	7	10	Lviv State University of Physical Culture, Lviv, Ukraine
Pimonenko, Tetyana	3	10	Sumy State University, Sumy, Ukraine	Female	12	23	Sumy State University, Sumy, Ukraine
Plastun, Alex	3	6	Sumy State University, Sumy, Ukraine	Male	0	1	Sumy State University, Sumy, Ukraine
Saik, Pavlo	3	6	Dnipro University of Technology, Dnipro, Ukraine	Male	2	4	Dnipro University of Technology, Dnipro, Ukraine; **Guangdong University of Petrochemical Technology, Maoming, China**
Shevchenko, Tetiana	3	6	Sumy National Agrarian University, Sumy, Ukraine	Female	2	2	Sumy National Agrarian University, Sumy, Ukraine; **Université Paris-Saclay, Gif-sur-Yvette, France**
Tkachenko, Victoria	3	3	Shupyk National Healthcare University of Ukraine	Female	0	0	Shupyk National Healthcare University of Ukraine
Yamelynets, Taras	3	4	Ivan Franko National University, Lviv, Ukraine	Male	0	0	Ivan Franko National University, Lviv, Ukraine

** Non-Ukrainian institutions are **bolded** in the table for clarity.*

Table 2. Authors affiliated with Ukrainian institutions with at least three publications in SSCI and AHCI in 2023

Authors	Papers in 2021		Affiliation in 2021	Gender	Papers in 2023		Affiliation in 2023
	SSCI & AHCI	WoS CC			SSCI & AHCI	WoS CC	
Artyukhov, Artem	0	7	Sumy State University, Sumy, Ukraine	Male	3	4	**University of Economics Bratislava, Bratislava, Slovakia;** Sumy State University, Sumy, Ukraine
Brik, Tymofii	0	2	Kyiv School of Economics, Kyiv, Ukraine	Male	4	6	Kyiv School of Economics, Kyiv, Ukraine
Chumachenko, Dmytro	0	1	National Aerospace University "Kharkiv Aviation Institute", Kharkiv, Ukraine	Male	3	8	National Aerospace University "Kharkiv Aviation Institute", Kharkiv, Ukraine
Danyliuk, Ivan	3	9	Taras Shevchenko National University of Kyiv, Kyiv, Ukraine	Male	5	7	Taras Shevchenko National University of Kyiv, Kyiv, Ukraine
Diachenko, Aleksandr	0	2	Institute of Archaeology, National Academy of Sciences of Ukraine, Ukraine	Male	4	6	Institute of Archaeology, National Academy of Sciences of Ukraine, Ukraine
Dluhopolskyi, Oleksandr	0	6	West Ukrainian National University, Ternopil, Ukraine; Ternopil Volodymyr Hnatiuk National Pedagogical University, Ternopil, Ukraine	Male	7	9	West Ukrainian National University, Ternopil, Ukraine; **WSEI University, Lublin, Poland***
Frankova, Iryna	0	1	Bogomolets National Medical University, Kyiv, Ukraine	Female	3	4	Bogomolets National Medical University, Kyiv, Ukraine; **Vrije Universiteit Amsterdam, Amsterdam, Netherlands**
Hutsol, Taras	0	4	Polissia National University, Zhytomyr, Ukraine	Male	5	15	Polissia National University, Zhytomyr, Ukraine
Kiriazova, Tetiana	6	9	Ukrainian Institute on Public Health Policy, Kyiv, Ukraine	Female	4	7	Ukrainian Institute on Public Health Policy, Kyiv, Ukraine
Kovalov, Bohdan	0	1	Sumy State University, Sumy, Ukraine	Male	3	4	Sumy State University, Sumy, Ukraine

Kozytska, Inna	4	4	Taras Shevchenko National University of Kyiv, Kyiv, Ukraine	Female	4	4	Taras Shevchenko National University of Kyiv, Ukraine
Krupelnytska, Liudmyla	0	1	Taras Shevchenko Natl Univ Kyiv, Kyiv, Ukraine	Female	3	3	Taras Shevchenko Natl Univ Kyiv, Kyiv, Ukraine
Kubatko, Oleksandr	0	3	Sumy State University, Sumy, Ukraine	Male	3	3	Sumy State University, Sumy, Ukraine
Kurapov, Anton	5	11	Taras Shevchenko National University of Kyiv, Kyiv, Ukraine	Male	8	8	Taras Shevchenko National University of Kyiv, Kyiv, Ukraine; **University of Salzburg, Austria**
Litvinova, Olena	0	0	National University of Pharmacy, Kharkiv, Ukraine	Female	3	6	National University of Pharmacy, Kharkiv, Ukraine; **Medical University of Vienna, Vienna, Austria**
Lyeonov, Serhiy	0	1	Sumy State University, Sumy, Ukraine	Male	6	10	Sumy State University, Sumy, Ukraine
Lyulyov, Oleksii	3	10	Sumy State University, Sumy, Ukraine	Male	13	25	Sumy State University, Sumy, Ukraine
Martsenkovskyi, Dmytro	1	1	Bogomolets National Medical University, Kyiv, Ukraine	Male	5	7	Bogomolets National Medical University, Kyiv, Ukraine
Meteliuk, Anna	0	2	International Charitable Foundation Alliance for Public Health, Kyiv, Ukraine	Female	3	5	International Charitable Foundation Alliance for Public Health, Kyiv, Ukraine
Mishchuk, Halyna	0	4	National University of Water and Environmental Engineering, Rivne, Ukraine	Female	4	7	National University of Water and Environmental Engineering, Rivne, Ukraine; **Pan-European University, Bratislava, Slovakia**
Nazarovets, Serhii	0	1	State Scientific & Technical Library of Ukraine, Kyiv, Ukraine	Male	4	5	Borys Grinchenko Kyiv University, Kyiv, Ukraine
Neyter, Roman	0	0	Kyiv School of Economics, Kyiv, Ukraine	Male	3	3	Kyiv School of Economics, Kyiv, Ukraine
Nosova, Bogdana	0	0	Taras Shevchenko National University of Kyiv, Kyiv, Ukraine	Female	3	4	Taras Shevchenko National University of Kyiv, Kyiv, Ukraine

Pavlova, Iuliia	6	9	Lviv State University of Physical Culture, Lviv, Ukraine	Female	7	9	Lviv State University of Physical Culture, Lviv, Ukraine
Petrytsa, Petro	0	0	Ternopil Volodymyr Hnatiuk National Pedagogical University, Ternopil, Ukraine	Male	3	4	Ternopil Volodymyr Hnatiuk National Pedagogical University, Ternopil, Ukraine
Pimonenko, Tetyana	3	10	Sumy State University, Sumy, Ukraine	Female	13	24	Sumy State University, Sumy, Ukraine
Polishchuk, Volodymyr	0	2	Uzhhorod National University, Uzhhorod, Ukraine	Male	6	6	Uzhhorod National University, Uzhhorod, Ukraine
Pykalo, Iryna	1	1	Ukrainian Institute of Public Health Policy, Kyiv, Ukraine	Female	4	5	Ukrainian Institute of Public Health Policy, Kyiv, Ukraine
Smaliychuk, Anatoliy	0	1	Ivan Franko National University of Lviv, Lviv, Ukraine	Male	3	3	Ivan Franko National University of Lviv, Lviv, Ukraine
Trach, Roman	0	7	**Warsaw University of Life Sciences, Warsaw, Poland**	Male	3	6	**Warsaw University of Life Sciences, Warsaw, Poland;** National University of Water and Environmental Engineering, Rivne, Ukraine
Trach, Yuliia	0	6	National University of Water and Environmental Engineering, Rivne, Ukraine; **Warsaw University of Life Sciences, Warsaw, Poland**	Female	4	6	National University of Water and Environmental Engineering, Rivne, Ukraine; **Warsaw University of Life Sciences, Warsaw, Poland**
Velykodna, Mariana	0	2	Kryvyi Rih State Pedagogical University, Kryvyi Rih, Ukraine	Female	3	5	Kryvyi Rih State Pedagogical University, Kryvyi Rih, Ukraine

** Non-Ukrainian institutions are **bolded** in the table for clarity.*